

Polyphony J. S. Bruna

PHD CANDIDATE · COGNITIVE AND INFORMATION SCIENCES

University of California, Merced

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Education

Research Interests: Neural Networks, Complex & Dynamical Systems, Collective Intelligence, Joint Action, Alignment

University of California, Merced

Merced, CA

PHD COGNITIVE AND INFORMATION SCIENCES

Expected 2027

- Committee: Christopher T. Kello (chair), Michael J. Spivey, and Jeff Yoshimi
- Master's thesis (*en route*): "Synergetic Mechanisms Underlying the Emergence of Alignment in Human Dyadic Interaction"

University of California, Los Angeles

Los Angeles, CA

GRADUATE FELLOW, INSTITUTE FOR PURE AND APPLIED MATHEMATICS

Fall 2024

- Program: Mathematics of Intelligences

Vassar College

Poughkeepsie, NY

BA COGNITIVE SCIENCE AND PHILOSOPHY (Minor: Japanese)

2018 - 2022

- Overall GPA: 3.99/4.00
- Honors: *Phi Beta Kappa*, *Sigma Xi*, *Psi Chi*, *cum laude generali et cum laude in materia subiecta*
- Thesis: "Leveraging Surprise in a Recurrent Neural Network to (de)Construct Morphological Complexity in Japanese" (Advisors: Jan Andrews and Josh de Leeuw)

Professional Experience

MEMBERSHIPS

- 2022-cont. **Phi Beta Kappa**, Academic Honor Society
- 2022-cont. **Sigma Xi**, The Scientific Research Honor Society
- 2021-cont. **Psi Chi**, International Honor Society in Psychology

POSITIONS

- 2024, 2026 **Humane Studies Fellow**, Institute for Humane Studies, George Mason University
- 2022-2023 **Adolph Sutro Fellow**, Vassar College
- 2021-2022 **Intern**, Cognitive Science, Vassar College
- 2021 **Senior Researcher**, Undergraduate Research Summer Institute, Vassar College
- 2019, 2020 **Research Fellow**, Undergraduate Research Summer Institute, Vassar College
- 2018-2021 **Research Assistant**, Cognitive Science, Vassar College

Publications

* Denotes co-principal authorship

IN PREPARATION

Bruna, P. & Gyllingberg, L. (2025). *Cognition without neurons: modeling anticipation in a basal reservoir computer*.
<https://arxiv.org/abs/2505.02114>

PUBLISHED

Kello, C., **Bruna, P.**, & Thao, K. (2026). Contextual assembly of lexical functions in large language models. *Behavior Research Methods* 58(19). <https://doi.org/10.3758/s13428-025-02898-7>

Bruna, P. and Kello, C. (2025). Least Effort and Alignment in Task-Oriented Communication. *Cognitive Science*, 49, Article e70062. <https://doi.org/10.1111/cogs.70062>

de Leeuw, J. R., Andrews, J., Barney, L., Bigler, M., **Bruna, P.**, Chen, Y., Cherry, R., Dowie, D. R., Forbes, E., Haffey, B., Hu, X., Jaklitsch, M., Leopold, N., Lewis, C., MacDonald, D., McShaffrey, C., Nakayama, K., Olstad, W., Peng, R., ... Zhang, L. (2021). Words May Jump-Start Meaning More Than Vision: A Non-Replication of Early ERP Effects in Boutonnet and Lupyan (2015). *Collabra: Psychology*, 7(1). <https://doi.org/10.1525/collabra.29763>

PROCEEDINGS

Bruna, P., Dale, R., Spivey, M., & Kello, C. (2026). Collective decision-making in coupled echo state networks. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 48. <https://escholarship.org/uc/item/4p27s372>

Nosrati, S., **Bruna, P.**, Spivey, M., & Marghetis, T. (2026). Spatial navigation at the extremes: The collective accomplishment of high-speed, high-risk navigation in Rally Racing. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 48. <https://escholarship.org/uc/item/0kr3d6pw>

Kello, C. & **Bruna, P.** (2024). Emergent Mental Lexicon Functions in ChatGPT. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46. <https://escholarship.org/uc/item/5m9098b5>

*Rane, S., ***Bruna, P.**, Sucholutsky, I., Kello, C., & Griffiths, T. (2023). Concept alignment. *1st NeurIPS Workshop on AI meets Moral Philosophy and Moral Psychology (MP2)*. <https://openreview.net/pdf?id=D3GCdfJL4J>

MISCELLANEOUS

Yoshimi, J., Beckmann, P., Udell, D., **Bruna, P.**, & Meyer, T. (2025). Transformer Architectures and LLMs. In Yoshimi, J., Udell, D., Tosi, Z., Hotton, S., Beckmann, P., Cain, E., Gordon, C., Noelle, D. C., **Bruna, P.**, & Meyer, T. (Eds.), *Neural Networks in Cognitive Science*. <https://downloads.jeffyoshimi.net/NeuralNetworksCogsci.pdf>

Abbas, M., Alharthi, D., Balbas, J., Beckelhymer, D., Beylkin, G., **Bruna, P.**, Cartmill, E., Chormai, P., Corea, F., Dai, X., Das, P., Dipoppa, M., Eberle, O., El Hady, A., Evans, J., Feghhi, E., Flack, J., Flores, C., Foster, J., ... Zhu, Y. (2025). White Paper: "Mathematics of Intelligences." *Institute for Pure and Applied Mathematics*. <https://www.ipam.ucla.edu/reports>

Bruna, P. (2019). A Review of Bryan W. Van Norden, Taking Back Philosophy. *Vassar College Journal of Philosophy*, 6. <https://www.vassar.edu/sites/default/files/2021-08/philosophy-journal-spring-2019.pdf>

Presentations

INVITED TALKS

Summer 2026. *Collective Decision-Making in Coupled Echo State Networks*. La Selva Conference on Language and Emergence. Santa Cruz, CA.

Spring 2026. *Collective Decision-Making in Coupled Echo State Networks*. Modeling the Social Mind(s): Theory-Based Computational Models (Symposium). The Society for Personality and Social Psychology Annual Convention, Chicago, IL.

Spring 2026. *Collective Decision-Making in Coupled Echo State Networks*. Social Interaction Lab, Dept. of Linguistics, Stanford University, CA.

Fall 2025. *Collective Decision-Making in Coupled Echo State Networks*. December Graduate Conference. Institute for Humane Studies, George Mason University, Fairfax, VA.

Fall 2024. *Least Effort and Alignment in Task-Oriented Communication*. Dept. of Cognitive Science, Vassar College, Poughkeepsie, NY.

CONTRIBUTED TALKS

Bruna, P. 2026. *Collective Decision-Making in Coupled Echo State Networks*. 48th Annual Conference of the Cognitive Science Society, Rio de Janeiro, Brazil.

Bruna, P. 2026. *Collective Decision-Making in Coupled Echo State Networks*. Mathematics of Intelligences Reunion Conference at Lake Arrowhead, Institute for Pure and Applied Mathematics, University of California, Los Angeles.

Bruna, P. 2025. *Collective Decision-Making in Coupled Echo State Networks*. Cognitive & Information Sciences Brownbag Series, University of California, Merced.

Bruna, P. 2025. *Collective Information Processing in Loosely Coupled Echo State Networks*. Graduate Workshop in Computational Social Science, Santa Fe Institute, New Mexico.

- Bruna, P.** 2025. *Basal Reservoir Computing*. Philosophy and Cognitive Science Working Group, University of California, Merced.
- Bruna, P.** & Gyllingberg, L. 2024. *Non-neural Pattern Completion in an Adaptive Reservoir Computer*. Mathematics of Intelligences Culminating Conference at Lake Arrowhead, Institute for Pure and Applied Mathematics, University of California, Los Angeles.
- Bruna, P.** 2024. *Least Effort and Alignment in Task-Oriented Communication*. Mathematics of Intelligences Seminar Series, Institute for Pure and Applied Mathematics, University of California, Los Angeles.
- Bruna, P.** 2024. *Concept Alignment*. Mathematics of Intelligences Working Group on Architecture and Evaluation of Generative AI, Institute for Pure and Applied Mathematics, University of California, Los Angeles.
- Bruna, P.** 2024. *Falandays-style Reservoir Networks*. Mathematics of Intelligences Working Group on Ecological and Homeostatic Intelligence, Institute for Pure and Applied Mathematics, University of California, Los Angeles.
- Bruna, P.** 2024. *Concept Alignment*. Philosophy and Cognitive Science Working Group, University of California, Merced.
- Bruna, P.** 2023. *Semantics and Syntax Co-emerge in Adaptive Reservoir Network Dynamics*. Diverse Intelligences Summit, University of St. Andrews, Scotland.
- Bruna, P.** 2020. *Hearing That It Is Silent: How to Hallucinate a Non-Perception*. 10th Annual Mid-Hudson Valley Undergraduate Philosophy Conference, Marist College, Poughkeepsie, NY. (Canceled due to COVID-19)

CONTRIBUTED POSTERS

* Presenting author(s); * Mentored undergraduate

- *Gaskin, B., **Bruna, P.**, Jackson, I., & O'Hearn, W. (2026). *BrainlessLab: From cellular cognition to collective intelligence*. The International Conference on Artificial Life, Waterloo, Ontario, Canada.
- *Nosrati, S., **Bruna, P.**, Spivey, M., & Marghetis, T. (2026). *Spatial navigation at the extremes: The collective accomplishment of high-speed, high-risk navigation in Rally Racing*. 48th Annual Conference of the Cognitive Science Society, Rio de Janeiro, Brazil.
- ***Bruna, P.**, Dale, R., Spivey, M., & Kello, C. 2025. *Coupled Echo State Networks as a Model of Task-Oriented Alignment*. UC Merced Cognitive Science Showcase, Merced, CA.
- ***Bruna, P.**, Dale, R., Spivey, M., & Kello, C. 2025. *Coupled Echo State Networks as a Model of Task-Oriented Alignment*. 47th Annual Conference of the Cognitive Science Society, San Francisco, CA.
- *Gyllingberg, L., Tian, Y., Sumpter, D., & **Bruna, P.** 2025. *Oscillations and Adaptation: Mathematical Models of Non-Neural Cognition*. Biological Systems that Learn, National Institute for Theory and Mathematics in Biology, Chicago, IL.
- ***Bruna, P.** & Kello, C. 2024. *Alignment Through Dyadic Least Effort in Task-Oriented Conversations*. 65th Annual Meeting of the Psychonomic Society, New York City, NY.
- ***Bruna, P.** & Kello, C. 2024. *Alignment Through Dyadic Least Effort in Task-Oriented Conversations*. Modeling Multi-Scale Collective Intelligences, Institute for Pure and Applied Mathematics, University of California, Los Angeles.
- *Kello, C. & ***Bruna, P.** 2024. *Emergent Mental Lexicon Functions in ChatGPT*. 46th Annual Conference of the Cognitive Science Society, Rotterdam, Netherlands.
- **Thao, K., **Bruna, P.**, & Kello, C. 2024. *Exploring Emergent Embodied Intelligence in Large Language Models: Insights from ChatGPT-4*. 15th Annual California Cognitive Science Conference, Berkeley, CA.
- *Rane, S. & ***Bruna, P.** 2023. *Concept alignment*. 1st NeurIPS Workshop on AI meets Moral Philosophy and Moral Psychology (MP2), New Orleans, LA.
- **Leshock, E., **Pena, Y., **Bruna, P.**, & Andrews, J. 2021. *Learned Categorical Perception with Artificially Generated Faces*. 36th Annual URSI Symposium, Vassar College, Poughkeepsie, NY.
- *Bigler, M. & ***Bruna, P.** 2020. *Probing the Methodology and Interpretation of Learned Categorical Perception Research*. 35th Annual URSI Symposium, Vassar College, Poughkeepsie, NY.
- ***Bruna, P.** & *Marks, P. 2019. *Psychophysical Adaptive Procedure: Developing a New, Generalizable Method*. 34th Annual URSI Symposium, Vassar College, Poughkeepsie, NY.

Honors

- 2025 **Best Poster Award - 2nd place**, *Cognitive Science Showcase, University of California, Merced*
Graduate Student Highlight ("Rising Star: How Life's Journey Shaped a Cognitive Scientist's Path to Discovery"),
<https://graduatedivision.ucmerced.edu/Polyphony-J-Bruna>
- 2022 **Best Technical Write-Up**, *Mid-Hudson IEEE Undergraduate Capstone Project Awards*

Awards, Fellowships, & Grants

** Awarded but declined*

2026	SSHA Dean Fellowship , <i>University of California, Merced</i>	\$ 27,300
	Residency Funding Award , <i>Institute for Humane Studies</i>	\$ 4,500
	Professional Development Award , <i>University of California, Merced</i>	\$ 1,725
	Graduate Student Travel Award , <i>University of California, Merced</i>	\$ 1,000
	Summer Research Fellowship , <i>University of California, Merced</i>	\$ 4,050
	Humane Studies Fellowship , <i>Institute for Humane Studies</i>	\$ 1,000
2025	Summer Research Fellowship , <i>University of California, Merced</i>	\$ 5,400
	Hayek Fund Travel Award , <i>Institute for Humane Studies</i>	\$ 3,475
	Graduate Workshop in Computational Social Science Scholarship , <i>Santa Fe Institute</i>	\$ 1,000
	Hayek Fund Travel Award , <i>Institute for Humane Studies</i>	\$ 2,400
2024	Humane Studies Fellowship , <i>Institute for Humane Studies</i>	\$ 5,000
	Fall Research Fellowship , <i>University of California, Merced</i>	\$ 6,000
	MacKenzie Scott Graduate Student Travel Award , <i>University of California, Merced</i>	\$ 1,000
	Travel Award , <i>University of California, Merced</i>	\$ 1,800
	Summer Research Fellowship , <i>University of California, Merced</i>	\$ 5,475
2023	Alumni Research Award , <i>Diverse Intelligences Summer Institute</i>	\$ 750
	Fall Research Fellowship , <i>University of California, Merced</i>	\$ 22,500
	Travel Award , <i>University of California, Merced</i>	\$ 800
	Summer Research Fellowship , <i>University of California, Merced</i>	\$ 1,600
2022	Chancellor Recruitment Fellowship , <i>University of California, Merced</i>	\$ 45,000
	*Foreign Language and Area Studies (FLAS) Fellowship , <i>U.S. Department of Education</i>	\$ 22,500
	Adolph Sutro Fellowship , <i>Vassar College</i>	\$ 10,000
	The Jamie Nisse Greenberg Philosophy Prize , <i>Vassar College</i>	\$ 900
	Critical Language Scholarship for advanced Japanese, alternate , <i>U.S. Department of State; American Councils for International Education</i>	
2021	Frances Lehman Loeb 1928 Scholarship , <i>Vassar College</i>	\$ 44,000
	Summer Research Fellowship , <i>Vassar College</i>	\$ 4,000
2020	Frances Lehman Loeb 1928 Scholarship , <i>Vassar College</i>	\$ 44,000
	Undergraduate Research Summer Institute Fellowship , <i>Vassar College</i>	\$ 5,000
2019	Frances Lehman Loeb 1928 Scholarship , <i>Vassar College</i>	\$ 42,500
	Undergraduate Research Summer Institute Fellowship , <i>Vassar College</i>	\$ 5,000
2018	1956 Memorial Scholarship , <i>Vassar College</i>	\$ 42,000

Teaching Experience _____

- Spr. 2026 **Complex Adaptive Systems**, Teaching Assistant, *Cognitive Science, University of California, Merced*
Fall 2025 **Mind, Brain, and Computation**, Teaching Assistant, *Cognitive Science, University of California, Merced*
Spr. 2025 **Critical Reasoning**, Teaching Assistant, *Philosophy, University of California, Merced*
Spr. 2024 **Contemporary Moral Problems**, Teaching Assistant, *Philosophy, University of California, Merced*
Spr. 2022 **Introduction to Cognitive Science**, Teaching Assistant, *Cognitive Science, Vassar College*
Fall 2021 **Introduction to Cognitive Science**, Teaching Assistant, *Cognitive Science, Vassar College*

Outreach & Professional Development _____

WORKSHOPS & TRAINING

- 2026 **Santa Fe Institute**, Advanced Graduate Workshop in Computational Social Science, *Santa Fe, New Mexico*
Diverse Intelligences Summer Institute, *Geneva, NY*
- 2025 **Santa Fe Institute**, Graduate Workshop in Computational Social Science, *Santa Fe, New Mexico*
Complexity Science Hub, Winter School on Complex Systems in Our Changing World, *Vienna, Austria*
- 2024 **Institute for Pure and Applied Mathematics**, Modeling Multi-Scale Collective Intelligences, *University of California, Los Angeles*
Berggruen Institute, Future Humans Workshop on Computational Consciousness, *Beverly Hills, California*
Institute for Pure and Applied Mathematics, Naturalistic Approaches to Artificial Intelligence, *University of California, Los Angeles*
Institute for Humane Studies, Aristotle and Human-Centered AI, *Webinar*
Institute for Pure and Applied Mathematics, Theory and Practice of Deep Learning, *University of California, Los Angeles*
Dynamic Interactions and Methodologies Symposium, *University of California, Los Angeles*
Institute for Pure and Applied Mathematics, Analyzing High-dimensional Traces of Intelligent Behavior, *University of California, Los Angeles*
- 2023 **Santa Fe Institute**, Complexity-GAINs International Summer School on Intelligence and Representation, *Isaac Newton Institute for Mathematical Sciences, University of Cambridge*
Diverse Intelligences Summer Institute, *University of St. Andrews*
- 2022 **Phi Beta Kappa**, Key Connections, *Webinar*
- 2020 **Center for Philosophy of Science**, Pittsburgh Summer Program in Philosophy of Science and Specialized Sciences, *University of Pittsburgh and Carnegie Mellon University*

SERVICE POSITIONS

- 2025-cont. **Book Club President**, Cognitive and Information Sciences, *University of California, Merced*
- 2025 **Brownbag Organizer**, Cognitive and Information Sciences, *University of California, Merced*
- 2023-2024 **Graduate Student Chair**, Cognitive and Information Sciences, *University of California, Merced*
Research Mentor, Learning Aligned Employment Program, *University of California, Merced*
- 2021 **Research Mentor**, Undergraduate Research Summer Institute, *Vassar College*

2018-2019 **Editor**, Vassar College Journal of Philosophy, *Vassar College*

PEER REVIEW

Cognition

Cognitive Science

Journal of Experimental Psychology: General

Minds and Machines

Proceedings of the Annual Meeting of the Cognitive Science Society

SKILLS

Programming: Python, R, JavaScript, HPC parallel computing

Open source software: crqa (R package, contributor)

Languages: English (native), Japanese (basic)